

RISHAV RAJ

AI/ML Engineer | LLM Fine-Tuning, Evaluation & Agent Security

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Summary

M.Sc. Computer Science graduate (2026) with hands-on work in LLM fine-tuning, held-out evaluation and failure analysis. Improved text-to-SQL execution accuracy from 50.6% to 68.4% on 21 held-out databases and tested an LLM agent against 100 live prompt-injection attacks. Thesis examined why TIES merging of distilled LoRA adapters behaved like plain averaging in the tested setup.

Skills

Languages & Tools: Python, SQL, Linux, Git, GitHub

LLM & ML: PyTorch, Hugging Face Transformers, PEFT/LoRA, TRL, GRPO, Unsloth, knowledge distillation, model merging (TIES), RAG, LangGraph

Evaluation & Data: execution-based evaluation, held-out and database-level splits, leakage checks, significance testing, adversarial testing, MLflow, embeddings, Pinecone

Serving & Cloud: vLLM, llama.cpp, GGUF quantization, FastAPI, Docker, AWS (EC2, Lambda, CloudWatch)

Projects

Text-to-SQL Fine-Tuning & Deployment | Qwen2.5-Coder-7B

Mar – Jun 2026

github.com/dLounce/Text-to-SQL | [Demo](#)

- Fine-tuned Qwen2.5-Coder-7B with LoRA SFT, then GRPO using SQL execution results as the reward. Execution accuracy rose from 50.6% to 68.4% on 1,600 questions from 21 databases held out of training.
- Prevented data leakage by splitting by database, so no schema appeared in both training and test. Removed examples whose gold SQL failed, timed out or returned no rows, and checked for exact duplicates, near-duplicates and overlap across splits. Built the GRPO set from prompts the SFT model solved in some, but not all, of 8 samples.
- Quantized the SFT model to GGUF Q4_K_M (15 GB to 4.4 GB, 0.6-point accuracy drop) and served it on AWS EC2 with llama.cpp and FastAPI, where a SQL-only request took about 12 seconds on CPU. Added input checks, a timeout, health checks and CloudWatch metrics; tracked four runs in MLflow.

Sentinel Procurement | LangGraph Agent Security

Aug – Sep 2026

github.com/dLounce/sentinel-procurement

- Built a procurement agent where the model reading untrusted vendor text has no tools, and the model placing orders sees only validated, typed offers. Budget, price and delivery checks run outside the models and must pass before an order can be created.
- Ran 100 live prompt-injection attacks across 10 attack families. 12 changed the buyer's decision beyond normal run-to-run variation, but none produced an unauthorized order or a policy violation.
- Compared four architecture variants in a 25-cell test matrix. All three weakened variants allowed unauthorized orders; the full design allowed none in these tests. 310 automated tests cover the checks, negotiation flow and evaluation code.

KD-TIES: TIES Merging of Distilled LoRA Adapters | M.Sc. Thesis

Dec 2025 – Jun 2026

github.com/dLounce/KD-ties

- Distilled Qwen2.5-32B into LoRA adapters for Qwen2.5-7B (instruction following and math, 5,000 samples each) and merged them with TIES. When adapters trained on the base model were merged into the instruct model, sign conflict was 0.000 and TIES behaved like plain averaging.
- Traced this to the instruction-tuning shift shared by every task vector. Subtracting it before merging (SC-TIES) restored sign conflict to 0.237 and produced the only cross-method gain above 2 standard errors (+0.017 MMLU).
- Reported the negative result: distillation had already reduced GSM8K from 0.632 to 0.466 before merging, and no tested merge variant recovered that loss.

Experience

Artificial Intelligence Intern | Ballistic Learning Systems, Faridabad, India

Jun – Jul 2025

- Evaluated an LLM grader against 118 human-labelled answers and improved agreement from 44.1% to 71.0% by adjusting the prompt and retrieved context.
- Built a RAG pipeline over course PDFs: chunked documents, stored embeddings in Pinecone and retrieved context for grading and question generation.

Education

M.Sc. in Computer Science

Jul 2024 – Jun 2026

Central University of Rajasthan, Ajmer | CGPA: 7.1/10

Bachelor of Computer Applications

Aug 2018 – Jul 2021

Patliputra University, Patna | 73%